🌐 Web Frameworks — Filtered & Organized Questions

🔵 UNIT 1 — PHP Fundamentals

1. **Explain the features of PHP.**
2. **Explain PHP variables and constants.**
3. **Explain variable scope in PHP with an example.**
4. **Explain operators in PHP.**
5. **Explain different conditional statements in PHP.**
6. **Explain `break` and `continue` statements.**
7. **Differentiate between `==` and `===` in PHP.**
8. **Explain the concept of classes in PHP.**
9. **Differentiate between `include` and `require` statements in PHP.**
10. **Explain different types of errors in PHP and methods of error handling.**
11. **Explain exception handling in PHP with `try`, `catch` and `throw` with an example.**

🔗 Related Topics

- Variables → Constants → Scope
- Operators → `==` vs `===`
- Errors → Error handling → Exceptions
- `break` → `continue`
- Class → Object-oriented PHP

🟢 UNIT 2 — PHP Arrays & String Handling

12. **Explain arrays in PHP and their types.**
13. **Explain multidimensional arrays in PHP with an example.**
14. **Explain array sorting functions in PHP.**
15. **Explain any three sorting functions used with arrays.**
16. **Write a PHP script to accept a sentence and count the number of words.**
17. **Explain string functions in PHP.**

> **Note:** The questions about "array creation" and "three sorting functions" are combined because they belong to the same array topic.

🟡 UNIT 3 — PHP File, Directory & Date-Time Handling

18. **Explain file and directory handling functions in PHP with suitable examples.**
19. **Explain date and time functions/methods in PHP with examples.**
20. **Explain file uploading in PHP.**
21. **Explain the PHP.INI configuration file.**

🔴 UNIT 4 — PHP State Management & Forms

22. **What is client-side and server-side state management? Explain the techniques.**
23. **What is a Cookie? Explain Cookies in PHP with an example.**
24. **What is a Session in PHP? Explain how sessions are created and handled with an example.**
25. **Differentiate between Cookie and Session.**
26. **Explain how to validate user input in PHP.**

⭐ Very Important Relationship

State Management → Cookie + Session → Cookie vs Session

Prepare these as one connected topic.

🟣 UNIT 5 — PHP & MySQL Database

27. **Explain the steps for connecting PHP with MySQL database.**
28. **Explain the interaction/communication between PHP and a database.**
29. **Explain database functions available in PHP.**
30. **Explain different types of tables in MySQL.**

🔗 Database Flow

PHP → Database Connection → SQL Query → Execute → Retrieve Data → Display Result

Questions 27 and 28 are essentially the **same core topic**, so one detailed answer can cover both.

🟤 UNIT 6 — PHP & Python Integration

31. **Explain methods of calling/executing a Python script from PHP with commands and examples.**
32. **Explain different techniques of calling PHP code from Python.**
33. **Explain the Python OS module and its functions/methods.**

> 🔴 **Important:** Python OS Module appears more than once, so prepare it thoroughly.

🟠 UNIT 7 — AJAX, JSON & Web Communication

34. **Explain AJAX request to PHP.**
35. **Explain how JSON is generated and parsed in PHP using `json_encode()` and `json_decode()`.**
36. **Explain the Request object and its uses.**
37. **Explain URL building with an example.**

🟦 UNIT 8 — CodeIgniter Framework

38. **Explain MVC architecture of CodeIgniter with a diagram.**
39. **Explain the purpose of URL Routing in CodeIgniter with an example.**
40. **Explain URL routing and route rules in CodeIgniter.**
41. **Explain the purpose of the Print/Template function with an example.**

> ⚠️ Some OCR wording in the original question bank is unclear, especially **"print template"** and the Flash/Flex question. I have kept the identifiable CodeIgniter topics but would avoid inventing the exact intended terminology.

🟩 UNIT 9 — Flash / Flex

42. **What is Flash? Explain how to create and host a Flash application.**
43. **Explain Flash/Flex application creation and uploading/hosting process.**
44. **Explain routes, route rules and URL handling in the Flash/Flex framework with suitable scripts.**

These questions appear to contain OCR errors, so this unit should ideally be checked against your **original syllabus or question-paper PDF** before preparing detailed answers.

📊 FINAL FILTERED QUESTION LIST

Unit	Topic	Questions
Unit 1	PHP Fundamentals	11
Unit 2	Arrays & Strings	6
Unit 3	Files, Directories & Date-Time	4
Unit 4	State Management & Forms	5
Unit 5	PHP + MySQL	4
Unit 6	PHP + Python	3
Unit 7	AJAX, JSON & Web Communication	4
Unit 8	CodeIgniter	4
Unit 9	Flash/Flex	3
	Total	**44**

🔥 Priority Based on Repetition

🔴 Priority 1 — Must Prepare

1. **PHP Variables, Constants & Scope**
2. **PHP Operators**
3. **Conditional Statements**
4. **Errors & Exception Handling**
5. **Arrays & Sorting Functions**
6. **Cookie**
7. **Session**
8. **Cookie vs Session**
9. **PHP + MySQL Database Connectivity**
10. **Python Script Execution from PHP**
11. **Python OS Module**
12. **File Uploading**
13. **CodeIgniter MVC**
14. **CodeIgniter URL Routing**

🟠 Priority 2 — Important

15. **PHP Features**
16. **Multidimensional Array**
17. **File & Directory Handling**
18. **Date & Time Functions**
19. **Input Validation**
20. **AJAX + PHP**
21. **JSON Encode/Decode**
22. **MySQL Table Types**
23. **Include vs Require**
24. **Classes in PHP**
25. **URL Building**

🟢 Priority 3 — Prepare Afterward

26. **`break` vs `continue`**
27. **`==` vs `===`**
28. **PHP.INI**
29. **Database Functions**
30. **Calling PHP from Python**
31. **Request Object**
32. **Flash/Flex topics**
